Welcome to !

Theory Of Automata

Introduction to languages

- There are two types of languages
  - Formal Languages (Syntactic languages)
  - Informal Languages (Semantic languages)

Alphabets

- Definition:

  A finite non-empty set of symbols (letters), is called an alphabet. It is denoted by Σ (Greek letter sigma).
- Example:

  Σ={a,b}

  Σ={0,1} //important as this is the language //which the computer understands.

  Σ={i,j,k}

NOTE:

- A certain version of language ALGOL has 113 letters

  Σ (alphabet) includes letters, digits and a variety of operators including sequential operators such as GOTO and IF

Strings

- Definition:

  Concatenation of finite symbols from the alphabet is called a string.
- Example:

  If Σ= {a,b} then

  a, abab, aaabb, abababababababababab

NOTE:

EMPTY STRING or NULL STRING

- Sometimes a string with no symbol at all is used, denoted by (Small Greek letter Lambda) λ or (Capital Greek letter Lambda) Λ, is called an empty string or null string.

  The capital lambda will mostly be used to denote the empty string, in further discussion.

Words

- Definition:

  Words are strings belonging to some language.

Example:

If Σ= {x} then a language L can be defined as

L={x^n : n=1,2,3,…..} or L={x,xx,xxx,….}

Here x,xx,… are the words of L

NOTE:

- All words are strings, but not all strings are words.

Valid/In-valid alphabets

- While defining an alphabet, an alphabet may contain letters consisting of group of symbols for example Σ_1= {B, aB, bab, d}.

- Now consider an alphabet Σ_2= {B, Ba, bab, d} and a string BababB.

This string can be tokenized in two different ways

- (Ba), (bab), (B)
- (B), (abab), (B)

Which shows that the second group cannot be identified as a string, defined over

$\Sigma = \{a, b\}$.

- As when this string is scanned by the compiler (Lexical Analyzer), first symbol B is identified as a letter belonging to Σ, while for the second letter the lexical analyzer would not be able to identify, so while defining an alphabet it should be kept in mind that ambiguity should not be created.

Remarks:

- While defining an alphabet of letters consisting of more than one symbols, no letter should be started with the letter of the same alphabet *i.e.* one letter should not be the prefix of another. However, a letter may be ended in the letter of same alphabet *i.e.* one letter may be the suffix of another.

Conclusion

- Σ_1= {B, aB, bab, d}
- Σ_2= {B, Ba, bab, d}

Σ_1 is a valid alphabet while Σ_2 is an in-valid alphabet.

Length of Strings

- Definition:

  The length of string s, denoted by |s|, is the number of letters in the string.
- Example:

  $\Sigma=\{a,b\}$

  s=ababa

  |s|=5

- Example:

  Σ= {B, aB, bab, d}

  s=BaBbabBd

  Tokenizing=(B), (aB), (bab), (d)

  |s|=4

Reverse of a String

- Definition:

  The reverse of a string s denoted by Rev(s) or s^r, is obtained by writing the letters of s in reverse order.

- Example:

  If s=abc is a string defined over Σ={a,b,c}

  then Rev(s) or s^r = cba

- Example:

  Σ= {B, aB, bab, d}

  s=BaBbabBd

  Rev(s)=dBbabaBB

Another Operation

$$w^n = \underbrace{www\cdots w}_{n}$$

$$(abba)^2 = abbaabba$$

Example:

$$w^0 = \lambda$$

Definition:

$$(abba)^0 = \lambda$$

The * Operation

Σ^* : the set of all possible strings from alphabet Σ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \square \}$$

The + Operation

Σ^+ : the set of all possible strings from alphabet Σ except λ

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \square\ \}$$

$$\Sigma^+ = \Sigma^* - \{\lambda\}$$

$$\Sigma^+ = \{a, b, aa, ab, ba, bb, aaa, aab, \square\ \}$$

Languages

A language is any subset of Σ^*

Example:

$$\Sigma = \{a, b\}$$

$$\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, \square\ \}$$

Languages:

$$\{\lambda\}$$

$$\{a, aa, aab\}$$

$$\{\lambda, abba, baba, aa, ab, aaaaaa\}$$

Note that:

Sets $\emptyset = \{\} \neq \{\lambda\}$

Set size $|\{\}| = |\emptyset| = 0$

Set size $|\{\lambda\}| = 1$

String length $|\lambda| = 0$

Another Example

$$L = \{a^n b^n : n \geq 0\}$$

An infinite language

λ

ab

$aabb$

$aaaaabbbbb$

$\in L$

$abb \notin L$

Operations on Languages

The usual set operations

$$\{a, ab, aaaa\} \cup \{bb, ab\} = \{a, ab, bb, aaaa\}$$

$$\{a, ab, aaaa\} \cap \{bb, ab\} = \{ab\}$$

$$\{a, ab, aaaa\} - \{bb, ab\} = \{a, aaaa\}$$

Complement:

$$\overline{L} = \Sigma^* - L$$

$$\overline{\{a, ba\}} = \{\lambda, b, aa, ab, bb, aaa, \ldots\}$$

Reverse

$$L^R = \{w^R : w \in L\}$$

Definition:

$$\{ab, aab, baba\}^R = \{ba, baa, abab\}$$

Examples:

$$L = \{a^n b^n : n \geq 0\}$$

$$L^R = \{b^n a^n : n \geq 0\}$$

Concatenation

$$L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$$

Definition:

$$\{a, ab, ba\}\{b, aa\}$$

Example:

$$= \{ab, aaa, abb, abaa, bab, baaa\}$$

Another Operation

Definition:

$$L^n = \underbrace{LL\cdots L}_{n}$$

$$\{a,b\}^3 = \{a,b\}\{a,b\}\{a,b\} =$$

$$\{aaa, aab, aba, abb, baa, bab, bba, bbb\}$$

$$L^0 = \{\lambda\}$$

Special case:

$$\{a, bba, aaa\}^0 = \{\lambda\}$$

More Examples

$$L = \{a^n b^n : n \geq 0\}$$

$$L^2 = \{a^n b^n a^m b^m : n, m \geq 0\}$$

$$aabbaaabbb \in L^2$$

Star-Closure (Kleene *)

$$L^* = L^0 \cup L^1 \cup L^2 \cup \cdots$$

Definition:

Example:

$$\{a, bb\}^* = \{\lambda, a, bb, aa, abb, bba, bbbb, aaa, aabb, abba, abbbb, \cdots\}$$

Positive Closure

$$L^+ = L^1 \cup L^2 \cup \cdots$$

Definition:

$$\{a, bb\}^+ = \left\{ \begin{array}{l} a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \cdots \end{array} \right\}$$

Defining Languages

- The languages can be defined in different ways , such as Descriptive definition, Recursive definition, using Regular Expressions(RE) and using Finite Automaton(FA) etc.

Descriptive definition of language:

The language is defined, describing the conditions imposed on its words.

- Example:

  The language L of strings of odd length, defined over Σ={a}, can be written as

  L={a, aaa, aaaaa,.....}
- Example:

  The language L of strings that does not start with a, defined over Σ={a,b,c}, can be written as

  L={b, c, ba, bb, bc, ca, cb, cc, ...}

- Example:

  The language L of strings of length 2, defined over Σ={0,1,2}, can be written as

  L={00, 01, 02,10, 11,12,20,21,22}
- Example:

  The language L of strings ending in 0, defined over Σ ={0,1}, can be written as

  L={0,00,10,000,010,100,110,…}

- Example: The language **EQUAL**, of strings with number of a's equal to number of b's, defined over Σ={a,b}, can be written as

  {Λ ,ab,aabb,abab,baba,abba,…}
- Example: The language **EVEN-EVEN**, of strings with even number of a's and even number of b's, defined over Σ={a,b}, can be written as

  {Λ, aa, bb, aaaa,aabb,abab, abba, baab, baba, bbaa, bbbb,…}

- Example: The language **INTEGER**, of strings defined over Σ={-,0,1,2,3,4,5,6,7,8,9}, can be written as

  INTEGER = {…,-2,-1,0,1,2,…}
- Example: The language **EVEN**, of stings defined over Σ={-,0,1,2,3,4,5,6,7,8,9}, can be written as

  EVEN = { …,-4,-2,0,2,4,…}

- Example: The language $\{a^n b^n\}$, of strings defined over $\Sigma=\{a,b\}$, as

  $\{a^n b^n : n=1,2,3,\ldots\}$, can be written as

  {ab, aabb, aaabbb,aaaabbbb,…}

- Example: The language $\{a^n b^n a^n\}$, of strings defined over $\Sigma=\{a,b\}$, as

  $\{a^n b^n a^n: n=1,2,3,\ldots\}$, can be written as

  {aba, aabbaa, aaabbbaaa,aaaabbbbaaaa,…}

- Example: The language **factorial**, of strings defined over Σ={1,2,3,4,5,6,7,8,9} *i.e.* {1,2,6,24,120,…}
- Example: The language **FACTORIAL**, of strings defined over Σ={a}, as {$a^{n!}$: n=1,2,3,…}, can be written as {a,aa,aaaaaa,…}. It is to be noted that the language FACTORIAL can be defined over any single letter alphabet.

- Example: The language **DOUBLEFACTORIAL**, of strings defined over Σ={a, b}, as

  $\{a^{n!}b^{n!} : n=1,2,3,\ldots\}$, can be written as

  {ab, aabb, aaaaaabbbbbb,…}
- Example: The language **SQUARE**, of strings defined over Σ={a}, as

  $\{a^{n_2} : n=1,2,3,\ldots\}$, can be written as

  {a, aaaa, aaaaaaaaa,…}

- Example: The language **DOUBLESQUARE**, of strings defined over $\Sigma=\{a,b\}$, as

  $\{a^{n^2} b^{n^2} : n=1,2,3,\ldots\}$, can be written as

  {ab, aaaabbbb, aaaaaaaaabbbbbbbbb,…}

- Example: The language **PRIME**, of strings defined over $\Sigma=\{a\}$, as

  $\{a^p : p \text{ is prime}\}$, can be written as

  {aa,aaa,aaaaa,aaaaaaa,aaaaaaaaaaa…}

An Important language

- **PALINDROME**:

  The language consisting of Λ and the strings s defined over Σ such that Rev(s)=s.

  It is to be denoted that the words of PALINDROME are called palindromes.

- Example:For Σ={a,b},

  PALINDROME={Λ , a, b, aa, bb, aaa, aba, bab, bbb, ...}

SummingUp Lecture-1

- Introduction to the course title, Formal and In-formal languages, Alphabets, Strings, Null string, Words, Valid and In-valid alphabets, length of a string, Reverse of a string, Defining languages, Descriptive definition of languages, EQUAL, EVEN-EVEN, INTEGER, EVEN, $\{ a^n b^n \}$, $\{ a^n b^n a^n \}$, factorial, FACTORIAL, DOUBLEFACTORIAL, SQUARE, DOUBLESQUARE, PRIME, PALINDROME.